

Question-1

(a) What are the limitations of Classical optimization problems? How Non Linear Programming (NLP) tries to overcome these problems?

(b) What is Kuhn-Tucker conditions (KTC)? Derive and interpret the KTC for a simple NLP.

(c) Math

Question-2

(a) “Cusps are the most frequently cited culprits for the failure of the Kuhn-Tucker conditions, but the truth is that the presence of a cusp is neither necessary nor sufficient to cause those conditions to fail at an optimal solution”. Explain this statement with appropriate example and show that the Constraint Qualification is satisfied if, for any point x^* on the boundary of the feasible region, there exists a qualifying arc for every test vector dx .

(b) Math

Question-3

(a) What is Maximum-value function? Evaluate the Envelop Theorem for both unconstrained optimization and constrained optimization.

(b) A consumer's expenditure function and his or her indirect utility function exemplify the minimum- and maximum-value functions for dual problems. Consider a consumer with the utility function $U(x,y) = xy$, who faces a budget constraint of B and is given prices P_x and P_y . Construct both Primal and Dual problems and show the duality.